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LLMs for legal reasoning: A unified framework and future perspectives

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ABSTRACT

Large Language Models (LLMs) have recently demonstrated remarkable ease of application to numerous natural language processing tasks, however the question of how well they perform is in serious question. In the case of their use in application domains where precision and accuracy are paramount (e.g., law, medicine), the assessment of their performance is erratic. In particular, the application of these models to legal reasoning presents both unique challenges and substantial opportunities because of the inherently complex and multi-faceted nature of legal decision-making. To begin to harness the potential of LLMs in legal reasoning, we propose a framework for unified legal reasoning that combines rule-based, abductive, and case-based approaches, and then investigate possible methods for their integration with LLMs. The ultimate goal, which we take steps toward, is to provide comprehensive, accurate, and adaptable legal decision analysis.

We critically examine this combination of reasoning methods, their formalizations, and their relevance to the legal domain, including the consideration of calibration methods to assess their performance. Moreover, we discuss current research and challenges in applying LLMs to legal reasoning tasks, highlight the importance of reconciling different reasoning paradigms, analyze cultural notions of justice, and address issues of uncertainty, vagueness, and ambiguity. Our study offers insights into the benefits and complexities of integrating LLMs within a proposed unified reasoning framework, with the hope of addressing some of the diverse legal challenges, and to advance the capabilities of AI-driven legal analysis.

1. Introduction

The intersection of Artificial Intelligence (AI) and the legal domain has experienced recent and notable advances, partially because of the rapid development and deployment of Large Language Models (LLMs) [1–3]. From early transformer models like BERT [4–6] to OpenAI's GPT series there has been rapid progress exemplified by unparalleled proficiency in numerous natural language processing (NLP) tasks largely related to the capture and processing of large volumes of legal texts [7–9].

The development trajectory of these models, in addition to the emergence of so-called "fine-tuning methods", has created a new LLM adaptation method called "prompt-engineering". Prompt engineering [10, 11] is basically the idea of elaborating LLM requests with subject matter expertise, to attempt to provide guidance on more practical and relevant LLM outputs. For example, the method of prompt engineering might be applied to restate a simple LLM request like "Crown versus Jordan" to a more elaborate request like "Show all relevant discussions regarding completion of judicial decision focused only on the Canadian Supreme Court of Canada". In principle, one expects *all* LLMs to produce more concise and focused output from the "re-engineered" second prompt. In general, prompt engineering is intended to provide a mechanism to compensate for the necessarily broad, often vague and typically uncontextualized "generative" responses from LLMs. The important challenge question is whether one can develop a framework that works in companion with generic LLMs to focus on accuracy and contextual relevant in legal reasoning.

Despite the foundational challenges of LLMs, there is a growing trend among legal technology companies to integrate LLMs into the

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[1] https://www.legaltechnologyhub.com/contents/the-use-of-large-language-models-in-legaltech dated February 18, 2023, authored by Nicola Shaver, CEO and co-founder of Legaltech Hub.

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services and solutions they offer. For example, a legal technology blog post,[1] provides a curated list of general legal office text processing tasks that can be supported by LLMs. This list highlights 27 companies that have publicly declared their integration of LLMs for specific use cases, across domains such as automatic generation of contract summaries, legal drafting, legal research, and the creation of a precedent database. While their breadth of application has been wide, the application of LLMs to challenges emerging from legal *reasoning* introduces a unique set of challenges and further opportunities.

The challenges are many: legal reasoning is inherently complex and multifaceted, involving the interpretation of laws, regulations, and precedents to support the construction of sound legal decisions. Traditionally, legal professionals and researchers have relied on rule-based, abductive, and case-based reasoning methods [12–14] to navigate the intricate landscape of legal analysis. These methods have been relatively effective, but typically require human expertise and intensive labor, further contributing to the time-consuming and resource-intensive nature of legal information processing. But LLMs have shown promise in legal text processing tasks, sometimes outperforming humans in accuracy, speed, and cost efficiency, especially for specific legal domains like contract review. While acknowledging that LLMs do not strictly reason in the conventional, human sense [15] and primarily operate on pattern completion, it would be also inappropriate to completely dismiss the reasoning capabilities of LLMs. Indeed, although it is true that LLMs function primarily by completing patterns and sequences, with few-shot prompting they do demonstrate some evidence of broader reasoning [16].

Upon encountering new patterns, LLMs have been incrementally improved to learn how to process and complete such new patterns based on the knowledge they already embed. For example, consider a scenario where an LLM is tasked with draft analysis in legal documents. By being prompted with a few examples of how certain clauses were interpreted in past cases, the LLM can predict potential interpretations [17] and flag sections that might need further review. This capability significantly reduces the review time while maintaining high accuracy, at least as measured by the coverage of identification of texts for further review.

Another example is observed in summarizing complex legal opinions [18]. An LLM, when trained on numerous legal documents, can produce concise and accurate summaries of legal opinions, making it easier for legal professionals to understand the core issues and rulings without wading through voluminous texts. Note that summaries are also resource saving for humans who have to digest large volumes of legal text.

Furthermore, LLMs have shown potential in automating the extraction of relevant precedents [19]. For example, given a legal query, an LLM can efficiently retrieve pertinent precedent cases, including summarizing the key findings and legal principles from those cases. This assists attorneys in quickly identifying relevant cases that support their legal arguments.

However, it is important to notice, for example, the incorrect outcomes from an LLM on this abductive task: imagine a case where the LLM is provided with a sequence of legal steps and is asked to determine the most likely preceding step. If the task involves determining the most likely scenario that led to a specific termination clause being invoked, an LLM might incorrectly infer that a breach was the cause, even though context from previous cases indicate an expiration of the contract term was more plausible.

Continuing on the general theme illustrated by this example, although LLMs cannot be deemed to perform what legal experts would consider as legal reasoning based on logical principles, there is some indication that LLMs can provide similar outcomes as one would expect from reasoning by searching for analogous patterns in their training data. Weikang Yuan et al. [20] also acknowledged this issue and proposed a multi-agent system called MALR to enhance an LLMs' legal reasoning capabilities. However, they ultimately admitted that their system still requires improvement and identified so-called "retrieval augmented generation" (RAG) as a promising approach. While we agree with their observation, we also believe that, beyond retrieving relevant precedents or legal statutes for a given legal query, the ability to apply reasoning to correctly use those findings in the current legal context is equally important.

We believe that observing and evaluating LLMs through the lens of classical reasoning frameworks – rather than treating them as black boxes that merely convert inputs to outputs – will allow us to better assess their effectiveness on specific reasoning tasks and provide a compass for guiding future improvements. This view can be further illustrated in the structural overview portrayed in Fig. 1.

The motivation behind our study lies in the potential transformative impact of integrating LLMs with legal reasoning processes. As AI technologies continue to evolve, there is a growing recognition of the need to leverage emerging advances to streamline and enhance legal analysis. LLMs, with their natural language understanding capabilities, have the potential to assist legal practitioners with support for comprehensive analysis of vast volumes of legal texts, statutes, and cases, with the goal of improving the efficiency and accuracy of legal decision-making—thus our motivation in combining the two.

The motivation further stems from the necessity to address the unique challenges posed by legal reasoning, which include ambiguity, uncertainty, and the nuanced nature of legal language. Unlike general NLP tasks, legal interpretation often involves resolving multi-layered ambiguity (e.g., "reasonable care" or "material breach"), reconciling conflicting precedents, and applying norms across diverse jurisdictions—tasks that exceed the capacity of rule-based or statistical NLP approaches [24,25]. Note it is often the case that this "ambiguity" can be intentional, for example, it may arise from intentional vagueness in legislative drafting, in order to provide for contextual consideration in judicial decision making.

Furthermore, legal decisions often carry high-stakes consequences, making precision and contextual understanding critical. Traditional legal analysis is time-consuming and error-prone due to the scale and complexity of legal materials. LLMs, trained on large-scale legal corpora, offer contextual reasoning and flexibility in interpretation. Empirical studies have shown that LLMs improve consistency and accuracy in tasks such as contract review, case classification, and legal summarization [26]. For example, ambiguous clauses in contract law or vague statutory terms in tort law often lead to varied judicial interpretations—LLMs can help expose such ambiguities and generate multiple interpretive paths, supporting human reasoning rather than replacing it.

Our study attempts to articulate one approach to the development of a unified reasoning framework that combines the strengths of rule-based, abductive, and case-based reasoning with the current natural language processing properties of LLMs. Our goal is to establish a comprehensive and versatile method for legal reasoning by leveraging the synergies of these distinct reasoning paradigms.

Rule-based reasoning is fundamentally a form of deductive reasoning [27,28], even though the mechanisms will also support abductive reasoning (e.g., see [29]). Deductive reasoning is about drawing specific conclusions by systematically applying explicit inference rules to particular cases or sets of facts. This method excels in situations where the problem domain can be precisely defined with explicit rules and conditions. Rules are explicit and transparent, making it easier to understand the reasoning process and outcomes. This is crucial, especially in applications where interpretability is essential. In the ultimate application in legal reasoning, the clarity of legal arguments emerges from the precision of the rule-based representations.

It should also be noted that, unlike the classical interpretation of rules as formal systems of logic (e.g., [30]), modern deductive reasoning systems also provide the foundations for so-called non-monotonic reasoning systems (e.g., [31]). Such systems employ

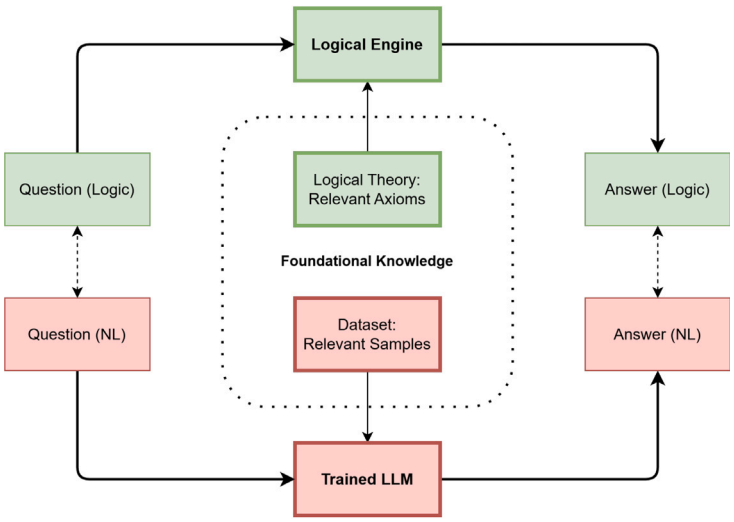


Fig. 1. An overview of similarities between logical (inference) engines and LLMs: both use appropriate knowledge to answer questions, with LLMs working directly on natural language and logical engines requiring conversion between representational forms. Some studies have been conducted on converting between these two forms of representation (e.g. by [21–23]).

deduction to reason about belief revision, which, in the context of legal reasoning, is about identifying the "best" sets of facts to support a decision.

Abductive reasoning is a foundational logical inference method for generating hypotheses or explanations for observed phenomena [32]. It is vital in legal reasoning as it helps legal professionals find plausible covering explanations where evidence and related legal cases have incomplete or ambiguous information. Generally, as noted above, the mechanism has been defined to exploit defeasible or non-monotonic reasoning with hypotheses as a foundation (e.g., [33]). Abductive reasoning also shares a foundation with *inductive reasoning*, as both can produce plausible but uncertain inferences and conclusions that are not deductively guaranteed. So inductive reasoning plays a significant role in legal settings, particularly in case-based reasoning. In that context, the flexibility and adaptability of abductive reasoning complements general inductive reasoning's pattern identification and generalization, allowing for more informed and well-rounded judgments in complex legal scenarios. Note that the original specification of inductive logic program by [34] and the specification of general first order abductive reasoning by [29,35] provide the foundation for combining deductive and abuctive reasoning.

Case-based reasoning is generally a framework that seeks to leverage past experiences and cases to solve new problems [36]. By storing and retrieving similar instances from a case base, the system can learn from historical data and apply knowledge gained in previous situations. Note that case reasoning in the legal domain is a kind of more specific instance of the general case of AI case-based reasoning, in the situation where judicial decisions rely on historical precedent. In the more general AI setting, the value of using previous solutions is the foundation of case-based reasoning, but the real foundational challenge is about how to identify what it means to be "similar", especially when the volume of previous cases is large, as in typical scenarios with legal case reasoning based on similarity of precedents. With this acknowledged, the general idea of case-based reasoning is that it exploits the consideration of the context in which a problem arises, thus enabling more nuanced and context-aware decision-making. As noted, the quality of general case-based reasoning vitally depends on common sense definitions of "similarity", (cf. [37]). Note further, especially considering our mandate to combine reasoning methods into a single framework, that the above mentioned inductive and abductive methods also provide the foundation for case-based reasoning, e.g., as noted in [38].

The **Unified Reasoning Framework** proposed here represents a new amalgamation of rule-based (deductive) reasoning, abductive reasoning, and case-based reasoning methodologies. This proposed framework intends to seamlessly integrate the unique strengths of these multiple reasoning paradigms, to create synergy and complementarity. This is intended to provide a dynamic and adaptive solution to intricate legal problem-solving scenarios, surpassing the limitations of each individual method. By incorporating rule-based reasoning, the framework provides a structured approach for well-defined problems, while abductive reasoning supports adaptation to uncertain or evolving situations by generating relevant plausible hypotheses, especially in the situation where information is dynamically accumulating, as is the case in the incremental gathering of evidence in legal reasoning. Furthermore, the inclusion of case-based reasoning enables the framework to exploit past experiences, especially in the form of argument structure, thus providing an overall comprehensive legal problem-solving approach that learns and improves over time. This versatility ensures the framework's efficacy in handling a diverse range of problem types, from rule-driven to complex and ambiguous scenarios. In essence, our framework stands as an integrative model, offering a holistic and cohesive solution to the challenges posed by the complex inferential processing required in legal reasoning. While the current framework focuses on deductive, abductive, and analogical or case reasoning for their formal tractability and integration potential, we recognize the importance of other reasoning modes such as policy-based and principled reasoning, and we address their roles and future inclusion in Section 6. Another timely research thread in the development of our integrated reasoning model is to consider the value and potential impact of LLMs. By integrating LLMs into our unified reasoning framework, we intend to establish a symbiotic relationship that leverages the strengths of both the human-like natural language-based reasoning of LLMs and the general value of machine learning. There are many potential advantages of such an integration for application in the legal domain, with the primary goal of advancing the performance of legal research and decision making systems. Note further, that this proposed combination of reasoning methods and LLMs is fully aligned with the most recent thrust of neuro-symbolic approaches to AI, which seek to generalize the foundation representation models of LLMs (e.g., [39]).

The rest of our paper is organized as follows. Section 2 delves into an illustrative case, examining potential legal reasoning approaches and their interactions in the resolution of a legal dispute. This example is intended to provide an intuitive basis for understanding the foundational challenges of legal reasoning, and for anticipating the potential value of our integrative model. Section 3 examines the major legal reasoning

approaches in the current artificial intelligence (AI) state of the art in legal reasoning. There is a lot of activity in this realm, and the challenge is to find an integrative thread that consolidates existing approaches to address the fundamental legal reasoning challenges. Section 4 outlines our proposed methodology for integrating LLMs into our unified reasoning framework. Following this, Section 5 discusses the proposed framework and assesses the effectiveness of that integration. Section 6 presents open issues and challenges in legal reasoning, emphasizing the alternative forms of reasoning to arrive at decisions. Finally, Section 7 concludes the paper.

2. An illustrative example of the legal reasoning challenge

Legal reasoning is shaped by the underlying legal framework within which judicial decisions are made. Broadly, legal systems can be categorized into two major traditions: **Civil Law** and **Common Law**. These systems differ in how legal rules are formulated, interpreted, and applied. Civil Law, which is prevalent in many jurisdictions (e.g., Japan. Europe), relies primarily on codified statutes, whereas Common Law places significant emphasis on judicial precedents (e.g., in Canada, originating in the UK).

Here, we illustrate essential aspects of legal reasoning in two existing major legal systems using a standard example case. The case is adapted from an actual case in the Courts of England and Wales in 1844 (*Young v. Hitchens*) and has been used as a demonstration in many publications on AI and law, including [40–43]. The case has the following facts.

Young v. Hitchens

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.
3. The plaintiff sued the defendant for theft.

First, let us consider Civil Law, which is adopted in much of the world. In a Civil Law legal system, judicial decisions mainly refer to codified legal rules, e.g., civil codes and criminal codes. Instances of Civil Law are mostly influenced by continental European countries like Italy, France, and Germany, so sometimes Civil Law is called Continental Law. Judges in Civil Law tend to employ proof structures for investigating the facts of the case and for drawing concluding legal decisions. Proof structures usually involve descriptions of wrong doings, such as guilt, and the burden of the proof usually starts from the plaintiff. Suppose that *Young vs. Hitchens* example case occurs in a Civil Law country. Then, the plaintiff needs to show, in the evidence associated with the claim, that the defendant committed the theft according to the proof structure. An example of a proof structure might be as follows.

(Fictional) Proof Structure

A person is guilty of theft if:

1. the person committed an action to take away an object
2. the object belonged to another person,
3. the person knew that the object belonged to the other person, and
4. the person had an intention to take away the object

except: the purpose of taking away is to prevent dangers (e.g., taking away weapons from a person that intends to injure someone).

The plaintiff might then claim and show evidence as follows.

Plaintiff Claim

The defendant is guilty of theft because

1. claim: the defendant committed an action to take away the fish
   evidence: the defendant caught the fish and rowed his boat out of the net with the fish
2. claim: the fish already belonged to the plaintiff
   evidence: the net was almost closed
3. claim: the defendant knew that the fish belonged to the plaintiff
   evidence: the defendant saw the net
4. claim: the defendant had an intention to take away the fish
   evidence: the defendant rowed his boat intentionally into the net

In turn, the defendant may argue that the fish did not belong to the plaintiff, as the net was not closed, but almost closed. If there are no statutes or previous cases that judges could follow, the judges need to decide whether we can claim that the fish belonged to the plaintiff, given that the net was almost closed. For example, the judges may invoke a criterion that the fish may be considered as belonging to the plaintiff only if the fish were attached to the net, or the net was a kind of dip net. Note that this subsumption between the concept of possession and the concept of catch can vary in time, culture, and contexts of each case.

Second, let us consider Common Law, which is adopted in many countries including the UK, the US, and Canada. In Common Law, judicial decisions mainly refer to past cases known as *precedents*. Judges in Common Law have the responsibility to interpret precedents and even distinguish or overturn precedents in certain circumstances, leading to the evolution of legal principles through cases. Plaintiffs and defendants in Common Law have to present and support their arguments using analogies from past cases. For example, consider two precedents to *Young v. Hitchens*, one supporting the plaintiff and another supporting the defendant, described below.

Keeble v. Hickeringill

1. The plaintiff owned land containing a pond, where the plaintiff always takes ducks for a profit.
2. The defendant, knowing about the decoy pond, fired the guns to scare away the ducks in the pond.
3. The court found in favor of the plaintiff as the plaintiff owned the pond.

Pierson v. Post

1. The plaintiff chased a fox in an open land
2. The defendant killed it and carried it away.
3. The court found in favor of the defendant as the defendant was hunting in an open land.

In the *Young vs. Hitchens* example case, the plaintiff may argue that he should win the case because he had made an effort to spread the net and catch the fish. He may support his argument using *Keeble v. Hickeringill*, where the pond is analogous to the area covered by the net that the plaintiff spreads, as both arise from that effort. In turn, the defendant may argue that he should win the case because fish in an open sea do not belong to anyone. He may support his argument using *Pierson v. Post*, where the fox in the open land is analogous to the fish in the open sea.

Judges and both plantiff and defendant parties have to interpret precedents by proposing *tests*, which are particular rules or rationales for decisions. The tests should follow the precedents as much as possible. For example, the judge may propose that *a wild animal belongs to the owner of the land where it lives; otherwise, a wild animal belongs to a person who can physically control it*. The test follows both precedents. Consequently, the defendant should win the case because the fish lived in an open sea and the defendant had controlled the fish physically. However, each party may propose alternative hypothetical cases to challenge the proposed tests [40].

Now further considering the example case, we can explore the possible legal reasoning approaches and how they might interact in supporting a legal judgement that resolves the dispute.

1. **Rule-based Reasoning:** From a rule-based perspective, we can investigate whether any statutes or regulations govern the ownership of fish in open waters. If such rules exist, the case can be decided accordingly. However, in the absence of clear-cut rules and evidence, it becomes necessary to explore other types of reasoning or their combination.
2. **Abductive Reasoning:** In applying abductive reasoning, we can hypothesize various explanations for the defendant's actions and the plaintiff's claims. For example, one hypothesis might be that the defendant entered the net unintentionally, or we could consider the possibility that this is a recurring issue between the plaintiff and the defendant. Evaluating these multiple hypotheses within the context of the given facts of the case can help us understand the underlying motivations and reach a fair decision.
3. **Case-based Reasoning:** Under case-based reasoning, we can try to find precedent cases that dealt with similar situations. Analogizing or distinguishing the current case from the precedents, we can examine how the courts applied the relevant legal principles and reach a consistent decision. The analogy requires the identification of sufficiently similar components of the precedents. In practice, judges might rely on this type of reasoning the most, given the importance of precedent in resolving disputes.

The combination of these reasoning methods underpins the complexity of legal decision-making tasks and demonstrates why relying on a single type of reasoning may be insufficient to resolve disputes. Integration of these approaches and managing their role in any case would provide a more comprehensive and nuanced analysis, leading to better-informed decisions.

3. AI aspects of major legal reasoning approaches

This section delves into the AI aspects of rule-based reasoning, abductive reasoning, and case-based reasoning within the legal domain, including both the challenges and the ongoing integration of LLMs into each legal reasoning.

3.1. Rule-based legal reasoning

Rule-based legal reasoning is the foundational form of deductive legal reasoning, where a legal conclusion is reached by applying general legal rules to specific factual circumstances. This reasoning mirrors the classical syllogistic logic structure, with legal rules as major premises, facts as minor premises, and conclusions as the result of applying the law to the facts. This interplay between legal norms and factual determination lies at the core of traditional judicial analysis. A typical rule-based legal system relies on three key components: (1) a set of legal rules derived from statutes, regulations, or precedent; (2) a body of facts specific to the case; and (3) an inference engine that systematically applies the rules to the facts to derive legal conclusions. In this deductive structure, once the facts are established, the legal outcome logically follows from the application of relevant rules. The structure of a syllogism can be expressed in the form of *major premise*, *minor premise*, and *conclusion*, such as the following:

Major Premise (Law): A person is guilty of theft if they intentionally take an object belonging to another person.
Minor Premise (Fact): Bob intentionally took Alice's purse.
Conclusion (Outcome): Bob is guilty of theft.

While this form of reasoning is conceptually straightforward, it demands precise articulation and formalization of both facts and legal rules. Historically, these tasks have required extensive manual labor from legal professionals [44]. With the rise of AI, and especially LLMs, there is now an opportunity to assist or automate parts of this process. LLMs have been applied both to extract legal rules from natural language and to simulate the reasoning process that applies those rules to factual inputs. Instead of relying on linguistically-oriented rule-based method [45], syntax-based approach [46], pattern matching techniques [47,48], or statistical model [49] for rule or fact extraction and formalization, transformer-based models such as BERT have emerged as a powerful alternative. In the COLIEE competition [50], most of the participating teams applied traditional information retrieval (IR) techniques such as BM25 [51], pre-trained language models such as BERT, or a combination of both to extract a subset of statutes $S1, S2, \ldots, Sn$ from the entire corpus of statutes (COLIEE uses the Japanese Civil Code,[2]) which are appropriate to answer the given legal question (COLIEE uses questions from the Japanese Bar Examination.[3]) In the task of determining whether a sentence describes facts or not, a meta-type system, developed by fine-tuning the RoBERTa base model, demonstrated superior performance compared to SVM models [52].

However, in the era of LLMs, machine-generated outputs also carry risks such as hallucination. This makes it crucial to integrate logical reasoning frameworks into human–AI collaboration workflows to mitigate systemic errors. For instance, no matter how advanced a user interface becomes, the decision to accept a terms-of-service agreement ultimately rests with the human; we cannot delegate that responsibility to an LLM by predicting the probability that a user would agree and binding them based on that estimate. Therefore, current research often focuses more on the extraction and formalization of legal norms rather than their application. For example, Liga and Robaldo [53] demonstrated that GPT-3 outperforms BERT, by achieving superior performance in the task of automatically detecting deontic rules from natural language. Janatian et al. [54] investigated whether a LLM is capable of analyzing legal rules and extracting a formal representation. They proposed and evaluated a framework that takes a piece of legislation (a single article or paragraph) and extracts a pathway of the rules contained therein, utilizing GPT-4, and achieved promising results.

3.2. Abductive legal reasoning

Abductive reasoning plays a vital role in legal reasoning as it involves forming and evaluating explanatory hypotheses. It is a form of imaginative and creative thinking that investigates possibilities and plausibilities based on the available evidence. Instead of deduction or induction, abductive reasoning proposes multiple possible explanations for a given phenomenon, which can then be evaluated against the evidence. This form of reasoning helps lawyers and legal scholars to articulate logical arguments, interpret laws, and develop legal theories [14].

Firstly, we consider the nature and characteristics of abductive reasoning. Abduction is different from deduction in that it does not show that something is necessarily true. Instead, it produces plausible hypotheses which can be tested against the available evidence [55]. Such reasoning can be layered, creative, revolutionary, and may involve visual and non-sentential aspects of hypothesis formation [56]. Furthermore, abductive reasoning in legal contexts is concerned with

[2] https://www.japaneselawtranslation.go.jp/en/laws/view/3494/en
[3] https://www.moj.go.jp/barexam.html

generating and discovering hypotheses that represent potential charges or complaints and relating them to evidence that can support or refute those hypotheses [14].

A variety of approaches have been proposed to address abductive reasoning in the legal domain. One approach uses logic programming, which offers a formal description of various methods for hypothesis generation and the evaluation of the best explanations among possible alternatives [55]. Abductive logic programming (ALP) extends traditional logic programming to include abduction as a form of reasoning and has found applications in knowledge assimilation, default reasoning, truth maintenance, and more [57].

Another approach to abductive reasoning is based on an argumentation-theoretic framework, which uses abductive inferences as interpretations for negation as failure. This approach links abduction with additional forms of explicit negation and provides a basis for the evaluation of explanatory hypotheses by means of argumentation [58].

The potential applications of abductive reasoning in legal contexts range from plan recognition and generation [55] to spotting legal issues in law school examination questions [58]. A more recent study proposed the idea of "double abduction" as an essential feature of legal reasoning, emphasizing that this process inherently involves a mechanism in which abductive inference is used to identify both potential legal categorizations and the relevant properties to be compared, thus providing support for a deeper understanding and analysis of the issue at hand through comparisons between two or more competing hypothesized interpretations of a phenomenon or situation specific evidence [59].

Despite the potential benefits of abductive reasoning, there are several challenges associated with its implementation in legal reasoning tasks. One of the main obstacles is the complexity of abduction, as the process can require a large number of possible hypotheses to be generated and evaluated. This can be computationally expensive and time-consuming for various legal domains [55]. Moreover, recent advancements in deep learning models have shown promise in legal text processing tasks, but they still struggle with abductive reasoning tasks, as demonstrated in a study using a logic-augmented dataset [60]. This highlights an area of improvement in the development of legal reasoning models to better support abductive reasoning.

More recently, researchers have investigated the capabilities of LLMs to perform abductive reasoning in various contexts [61]. For instance, uncommonsense abductive reasoning focuses on generating natural language explanations for unexpected outcomes in unusual situations. In this regard, a new English language corpus called UNcommonsense [62] was curated, which examines and compares the performance of human explainers and LLMs in generating explanations for uncommon situations.

Another study demonstrated the potential of LLMs in automating the credit risk analysis process by proposing a novel prompt-engineering method called Labeled Guide Prompting (LGP) [61]. LGP enhances the ability of LLMs to generate reliable credit risk reports by augmenting abductive reasoning with annotated few-shot examples and information about the interactions between variables from a Bayesian network used for credit risk assessment. Those experimental results showed that LLMs can generate credit risk reports preferred by human credit analysts and significantly improve the insightfulness of their responses.

However, there is evidence that LLMs, such as GPT-3 and GPT-4, still struggle to achieve high performance in deep abductive reasoning tasks [63]. A study introduced a challenging benchmark of long-form mystery narratives constructed as detective puzzles, which demonstrated that LLMs performance is barely beyond random, and there remains a significant gap between LLMs and human reasoning abilities. This calls for further research to improve the deep reasoning capabilities of LLMs.

Moreover, LLMs can improve event prediction by few-shot abductive reasoning, as shown in a study that developed a framework called LAMP for integrating LLMs in event prediction tasks [64]. Through abductive reasoning, LLMs can suggest possible causes for future event predictions based on past events, thus significantly outperforming state-of-the-art event sequence models in various real-world datasets.

In summary, abductive reasoning is an essential component of legal reasoning, and involves the formation and evaluation of explanatory hypotheses. In legal contexts, various challenges remain, such as comprehending and representing intricate legal concepts, managing conflicting evidence, and producing contextually relevant and logically consistent explanations. While various approaches have been explored to formalize and implement abductive reasoning in legal contexts, challenges persist in terms of the complexity of the reasoning process and the limitations of existing legal reasoning models. Despite its application in areas such as uncommonsense reasoning, credit risk analysis, and event prediction, current LLMs still face difficulties in deep abductive reasoning tasks. The limited performance of these models on reasoning benchmarks [60,65] highlights the need for further advancements to overcome these difficulties and enhance the effectiveness of abductive reasoning in legal domains.

3.3. Case-based legal reasoning

Case-based reasoning is a form of reasoning commonly used in law, particularly in Common Law. Even in Civil Law, judges often cite precedents to discuss their consistent application of laws [66]. According to [67], decisions in Common Law usually consist of four components: facts (situations that occurred in the case), issues (questions in the case that have to be resolved), reasoning (appropriate resolutions of the issues; sometimes called rationales), and outcomes (results of the case). Consider the example case *Young v. Hitchens* (see Section 2), where the issue was the ownership of the fish. The outcome of the case is for the defendant with the reasoning that the plaintiff was not the owner of the sea and the defendant had controlled the fish physically. To reach a decision, each side of the dispute (plaintiff/defendant) has to cite precedents that favor their side and are relevant to the facts of the present case. This process of citing precedents creates a web of interconnected cases, where each new decision builds upon and clarifies the principles established in those precedents. This reliance on precedents is known as *stare decisis*. Meanwhile, Common Law also allows for distinguishing, overruling, or refining precedents under certain circumstances. Ultimately, the goal is to find a just and equitable outcome in the present case while maintaining the overall coherence of the legal system.

Law has been a prime testing ground for case-based reasoning systems, leading to the development of a variety of particular methods. For example, following the structure of legal cases, case-based reasoning systems typically represent a case using a predetermined set of abstract features, such as a set of dimensions in HYPO [68] or a set of factors in CATO [69]. Those features are selected by human experts to capture the facts of the cases and how those facts strengthen or weaken the position of each party on an issue/outcome. Similarly, a variety of approaches have been introduced to draw conclusions: for example, HYPO introduces "claim lattices" to determine *on-pointness*, implying that an analogy between two cases can be identified by looking at the dimensions they have in common. Similarly, CATO introduces "factor hierarchies", which group factors based on legal issues, and then provides a reasoning process based on factor hierarchies. In another approach, IBP [70] introduces logical models of issues to provide predictions and explanations. In addition, "precedential constraints" [71] introduces a reason set, which is a subset of the identified facts which can be confirmed to dominate other factors and which favors the outcome decided. The idea of precedential constraints enforce a new decision to prevent introduction of a new inconsistency in a priority induced by reason sets. This idea is similar to the long tradition of argumentation theory, where an argument

is defined in terms of supporters and defeaters (e.g. [72,73]). Furthermore, researchers explore argumentation-theoretic frameworks for case-based reasoning, for example, abstract dialectical frameworks [74] and abstract argumentation for case-based reasoning [75].

Recent progress in NLP, particularly with transformer methods to build LLMs, have opened up exciting possibilities for case-based reasoning in law. While some studies, like [76], focus on predicting case outcomes directly from case descriptions, the AI and Law community is more interested in understanding the underlying factors and reasoning within legal cases. In addition to explanability, this interest relies on a key difference between legal reasoning and typical machine learning tasks: the ultimate goal of legal case-based reasoning is not a single, generalizable model, but rather a web of interconnected cases built through citations. This contrasts with LLMs, which are designed to generalize rather than memorize examples [77]. Relying solely on LLMs for case-based reasoning can generate hallucinations of interconnected cases, including the citing of fabricated non-existent cases.

As a consequence, some researchers are exploring hybrid approaches [78,79] that combine NLP techniques with traditional case-based reasoning systems. As noted above, the hybrid approach is beginning to exploit the emerging frameworks arising from the development of neuro-symbolic frameworks (e.g., [39]). This strategy involves using machine learning or LLMs for tasks like case retrieval and case entailment, followed by more traditional case-based reasoning systems to analyze and predict an outcome for the present case. This interest has led to machine learning competitions like COLIEE [50]. COLIEE Task 1 exemplifies the case retrieval task, requiring participants to find supporting cases for a new case from an entire corpus. This has led to further research on encoded summarization techniques for efficient case retrieval [80]. For example, COLIEE Task 2 exemplifies the case entailment task, by requiring participants to identify which paragraph in a relevant case entails a specific decision. Further research has explored other meaningful components, such as [81], who identify particular sentences in precedent for argumentation about the meaning of statutory terms. Additionally, recent studies like [82,83] explore the learnability of argumentation-theoretic frameworks for case-based reasoning. In general, research on argument identification and reasoning continues, and competition datasets like COLIEE help support this work.

4. Unified reasoning framework with LLMs

In this section we propose a unified reasoning framework which combines rule-based reasoning, abductive reasoning, and case-based reasoning with large language models (LLMs). We provide a detailed analysis of how these reasoning approaches can be unified and integrated with LLMs, and how the unified framework can simulate court procedures.

4.1. Components of the framework

The framework considers four components: fact, theory, hypothesis, and outcome, as illustrated in Fig. 2. The fact component provides the context of the case. The theory component provides applicable and defeasible rules derived from statutes or precedent cases. The hypothesis component provides hypothetical facts and rules derived from the context, including those with less certainty (e.g., intentions and motivations). Finally, the outcome component provides the proposition on specific issues (e.g., guilty/not guilty). The interplay between these components can simulate the three main reasoning modes as follows.

1. **Rule-based Reasoning:** Rule-based reasoning can be simulated by considering the components of fact and theory as the main input. The framework evaluates the plausibility of the outcome from fact and theory. Since theory is defeasible and factual description alone are often incomplete or ambiguous, the user can assert hypothetical rules or facts from the hypothesis component that the user accepts to challenge the reasoning in this mode. The hypothesis component can help bridging the gap between the known facts and the open texture of the legal rule (e.g., intentions).
2. **Abductive Reasoning:** Abductive reasoning can be simulated by considering the components of fact, theory, and outcome as input. The framework evaluates the plausibility of the hypothesis as an explanation for why the outcome can be deduced from the fact with respect to the theory.
3. **Case-based Reasoning:** Case-based reasoning can be simulated by considering the components of fact (provided the context of the present case and the precedent case) and outcome as input. The framework evaluates the plausibility of two components. The first component is the theory – explaining why both cases can deduce the same outcome. The second component is the hypothesis – providing analogical explanations between the cases and the theory.

4.2. Approaches to integrating LLMs

LLMs can be integrated in our unified framework through several architectural approaches, each offering unique benefits and limitations. One common approach involves using pre-trained LLMs, such as GPT, Claude, Llama or Gemini, which are general-purpose models that can be applied to legal contexts without significant modifications. These models can approximate some forms of legal reasoning but may lack domain-specific precision due to their broad training datasets.

Another advanced method is Retrieval-Augmented Generation (RAG), which enhances LLM performance by incorporating legal document retrieval into the response generation process. This technique allows models to identify relevant statutes, case law, or regulations in external databases, to help ensure that overall responses are supported by authoritative sources, rather than relying solely on the model's internal general knowledge. By integrating this form of retrieval mechanism, RAG reduces the risk of hallucinations and improves the factual accuracy of legal reasoning.

A more specialized approach is to "fine-tune" LLMs, which is a process whereby models are further trained on legal-specific datasets to augment and enhance their understanding of legal terminology, precedents, and statutory interpretation. The goal of the approach is to utilize the logit values from the final layer of an LLM to represent the relationship between the four components of the unified reasoning framework. The input to the LLM will consist of a sequence S, of concatenated text components—fact (F), theory (T), hypothesis (H), and outcome (O). The LLM will process this input sequence and generate an output for each token:

$$S = Encode(F, T, H, O)$$

We aim to extract a scalar value (logit) from the final layer output of the LLM that indicates the strength of the relationship between the premises, conclusions, and theory. This can be done by applying an aggregation function (Agg) on the neuron activations of the final layer, denoted as O_i, corresponding to a specific token or set of tokens.

Let $O(S)$ denote the final layer output (before $Softmax$) for the input sequence S in the LLM. We can calculate the logit value by aggregating output neuron activations using the aggregation function:

$$Logit = Agg(O(S)).$$

This logit value will be used to indicate the plausibility of the given theory concerning the premises and conclusions. A higher logit value signifies a stronger relationship between fact, theory, hypothesis, and goal. Conversely, a lower logit value indicates a weaker relationship. We note that this heuristic ranking of the logit relationship amongst the four components of the unified reasoning framework provides a kind of decision support for judicial decision making, but whose accuracy

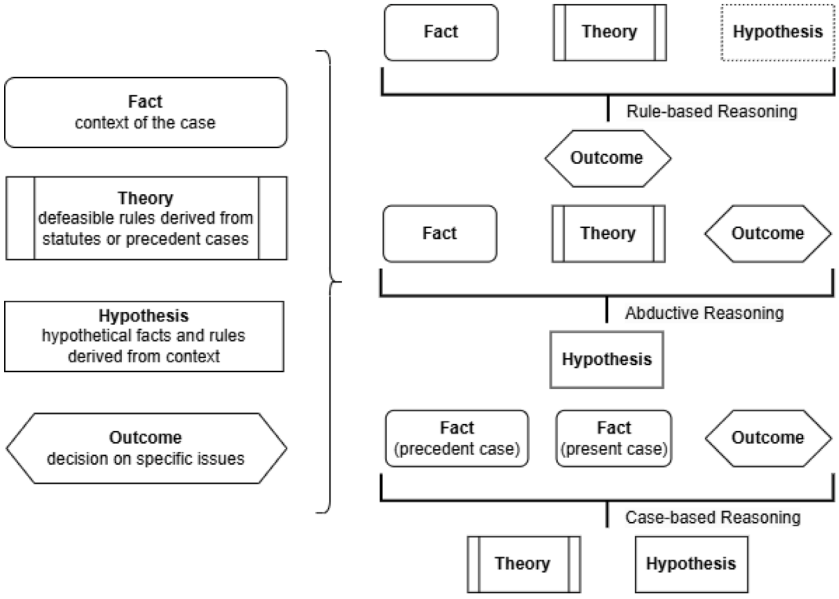


Fig. 2. Unified reasoning framework components.

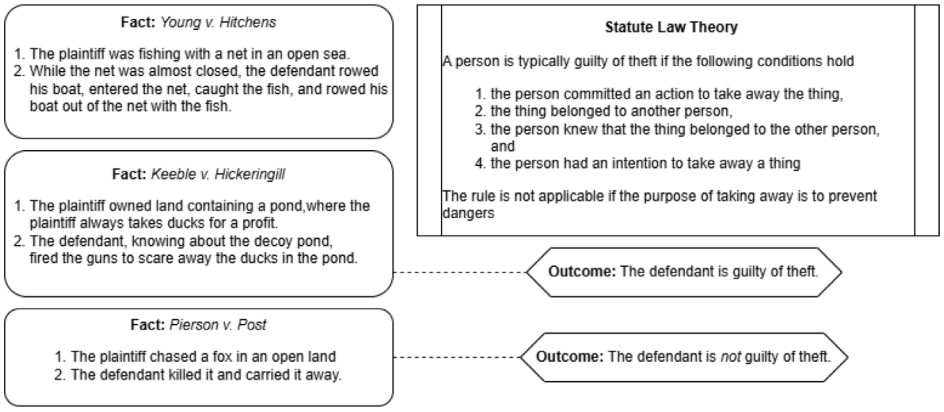


Fig. 3. Initial components from the example.

depends on the quality of each of the four components as well as on the quality of the corpora of training instances.

With this approach, and the goal to learn a suitably augmented LLM for the unified reasoning framework, we need to provide training examples with appropriate input sequences and label information. The training set should include various reasoning scenarios involving rule-based, abductive, and case-based reasoning with varying levels of associativity between premises, conclusions, and theories.

During this fine-tuning augmentation, the model will optimize its internal representations to minimize the difference between its logit predictions and the ground truth labels provided for each training example. After the fine-tuning process, the augmented LLM can be applied to various inference tasks in the unified legal reasoning framework using the input representation and logit outputs detailed above.

4.3. Practical legal reasoning example

To illustrate how the LLM-integrated framework simulates court procedures, we apply the example cases and rules in Section 2 and use Gemini-1.5-flash, which is a pre-trained LLM provided by Google. The initial components of the example cases can be illustrated as in Fig. 3. The example prompts and responses are described in detail in Appendices A and B.

To simulate a civil law court procedure, the framework starts from the rule-based reasoning mode. In this mode, an LLM is prompted by the facts of the case and the relevant legal theory, and asked to evaluate the plausibility of the outcome A.1. If either the plaintiff or the defendant is not satisfied with the outcome, they can re-prompt the LLM in abductive reasoning mode to generate hypotheses that support their intended outcome A.2. The judge then has the discretion to accept or reject these abductive hypotheses, refine the facts, or revise the theory, as required. If a new hypothesis is accepted, the LLM is again prompted with the new hypothesis in rule-based reasoning mode to reassess a new outcome A.3. This process is then repeated from the perspective of the opposing side. By iteratively refining these components, the framework aims to converge on a set of facts, theory, and hypothesis, which together form the argument structure that explains the reasoning behind the decided outcome.

To simulate a common law court procedure, the framework starts in the case-based reasoning mode. In this mode, either the plaintiff or the defendant provides two fact components: one describing the context of the present case and the other describing a relevant precedent case that supports an outcome in their favor. An LLM is prompted to generate a legal theory and an explanatory hypothesis for why the current case might have the same outcome as the precedent case B.1.

In response, each side can challenge the theory from the opposing side in two ways: (1) by proposing their explanatory hypothesis

generated from the case-based reasoning mode or, (2) by introducing a new hypothesis using the abductive reasoning mode B.2. The judge then has the discretion to accept or reject these hypotheses, refine the facts, or revise the theory as required. If a new hypothesis is accepted, the LLM is prompted again, this time in rule-based reasoning mode, to reassess the outcome from each side's perspective based on the revised hypothesis B.3. The framework aims to converge on a single outcome, determined by a settled fact, legal theory, and explanatory hypothesis, which together provide a structured rationale for the court's decision.

4.4. Experimental evaluation

To assess the effectiveness of the proposed unified reasoning framework when integrated with contemporary large language models (LLMs), we conducted a structured evaluation involving multiple prominent models across the three reasoning modes: rule-based, abductive, and case-based. The evaluation consists of 8 task prompts in total – 3 rule-based, 3 abductive, and 2 case-based – carefully designed following the component structure defined in Section 4 (see Appendices A and B).

We adopt the "LLM-as-a-Judge" framework [84], in which strong LLMs such as `gpt-4o`, `claude-3-5-sonnet`, and `gemini-1.5-pro` act as expert evaluators. These judge models scored each response on five dimensions: reasonableness, completeness, clarity, logical structure, and adherence to the legal reasoning framework. Each criterion is rated from 1 to 5, and the overall score is computed as the average. Below is the prompt used by judges:

Prompt

You are a legal expert evaluating AI model responses for legal reasoning quality.
ORIGINAL PROMPT: {*original_prompt*}
RESPONSE TO EVALUATE: {*response_to_evaluate*}
EVALUATION CRITERIA:

1. **Reasonableness**: Logic and soundness of legal reasoning
2. **Completeness**: Addresses all relevant factors and conditions
3. **Clarity**: Clear and understandable presentation
4. **Logical Structure**: Coherent chain of reasoning
5. **Framework Adherence**: Follows required legal reasoning framework

RESPONSE FORMAT: Respond only in JSON format:

```
{
  "reasonableness": {"score": [1-5], "explanation": "Brief explanation"},
  "completeness": {"score": [1-5], "explanation": "Brief explanation"},
  "clarity": {"score": [1-5], "explanation": "Brief explanation"},
  "logical_structure": {"score": [1-5], "explanation": "Brief explanation"},
  "framework_adherence": {"score": [1-5], "explanation": "Brief explanation"},
  "overall_score": [average score],
  "overall_comment": "Overall assessment of the response"
}
```

As shown in Fig. 4, flagship models such as the `gemini-1.5-pro`, the `claude-3-5-sonnet`, and the `gpt-4o` consistently achieved the highest overall scores across all prompts, with average scores above 4.8 out of 5. These models demonstrated strong coherence, complete reasoning chains, and adherence to the legal reasoning framework, even in abductive and analogical tasks that demand more interpretative inference. In contrast, standard-tier models like `gpt-3.5-turbo` and `gemini-1.5-flash` showed significantly more variability and lower alignment with the expected legal reasoning structure, particularly in prompts requiring multi-step or hypothesis-driven reasoning.

To further understand these differences across individual evaluation criteria, we present a heatmap in Fig. 5 showing the average score of each model on the five core dimensions: reasonableness, completeness, clarity, logical structure, and framework adherence. The heatmap reveals that flagship models not only score highly overall, but also maintain balanced performance across all criteria. In contrast, some standard models perform well in clarity but fall short in framework adherence or logical structure, indicating a less systematic reasoning process.

Together, these results support the claim that high-capacity LLMs are substantially more capable of executing structured legal reasoning when integrated with our unified framework. They also highlight the importance of both model quality and framework alignment in developing reliable AI-assisted legal tools.

5. Discussion

In our unified framework, LLMs participate in simulating legal inference based on a fact pattern, a rule pattern (theory) and possibly a hypothesis that interprets ambiguous elements. The framework evaluates the plausibility of different legal outcomes. This triadic interaction allows the framework to test how well rules connect with facts and which interpretations are most plausible under specific circumstances. For example, in *Young v. Hitchens*, the legal question was whether the fish, caught in an almost-closed net, were already the property of the plaintiff. The rule states that theft occurs if someone intentionally takes an object that belongs to another. The model must assess whether the condition"belonging to another" is satisfied, which depends on how the facts ("almost closed net") are interpreted.

Researchers have long been interested in unifying various reasoning approaches to support legal inference, such as [38,40]. Our proposed framework pioneers this unified approach with mechanisms coupled with LLMs. To enable this, we need to adapt several features from previous work, which typically operate within the realm of formal logic and ontology. For instance, we follow [38] in using abductive methods for analogical reasoning and we follow [40] in using precedent and hypothetical cases to challenge the current legal theory. Even with this approach, we may still loose some features of these formal methods, such as confirmation of hypothesis consistency and full resolution of theory disambiguation, as they might not be computationally feasible when dealing with the complexities of natural language and the nuances of legal arguments. However, we foresee that unifying these three reasoning approaches and integrating them with LLMs offers significant advantages:

- **Complementarity**: Rule-based, abductive, and case-based reasoning provide complementary approaches to legal challenges, each with their strengths and weaknesses. Unifying these approaches creates a comprehensive legal reasoning system that leverages the benefits of each individual method, while addressing their limitations.
- **Joint Representation**: All three reasoning methods involve relationships between premises and conclusions. By representing legal cases, rules, and analogies within a joint framework, we can develop a coherent representation that captures the essence of all reasoning approaches, and supports explainability.
- **Machine Learning Integration**: Integrating rule-based, abductive, and case-based reasoning within a unified framework provides structured input for machine learning algorithms, particularly the transformer methods used to construct LLMs. This makes legal reasoning tasks more amenable to neuro-symbolic AI models, and enables improved representation and prediction capabilities.

These benefits underscore the potential of integrating rule-based, abductive, and case-based reasoning with LLMs as part of a unified reasoning framework. As we argued, this approach can significantly enhance the performance and utility of AI models in addressing complex legal challenges, and can provide valuable insights and support for legal professionals across a wide array of legal scenarios. However, applying this unified reasoning framework with LLMs to real-life legal scenarios is not straightforward because of their current limitations, as discussed in previous sections; for example, there remain challenges with accurate comprehension and representation of legal language, continued struggles with deep abductive reasoning, and the need for

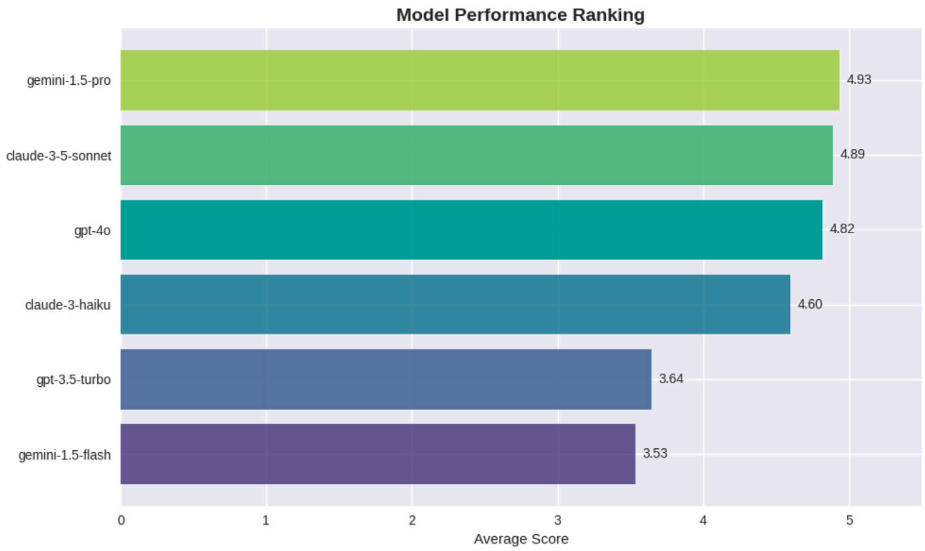


Fig. 4. Overall performance ranking of LLMs across unified legal reasoning tasks.

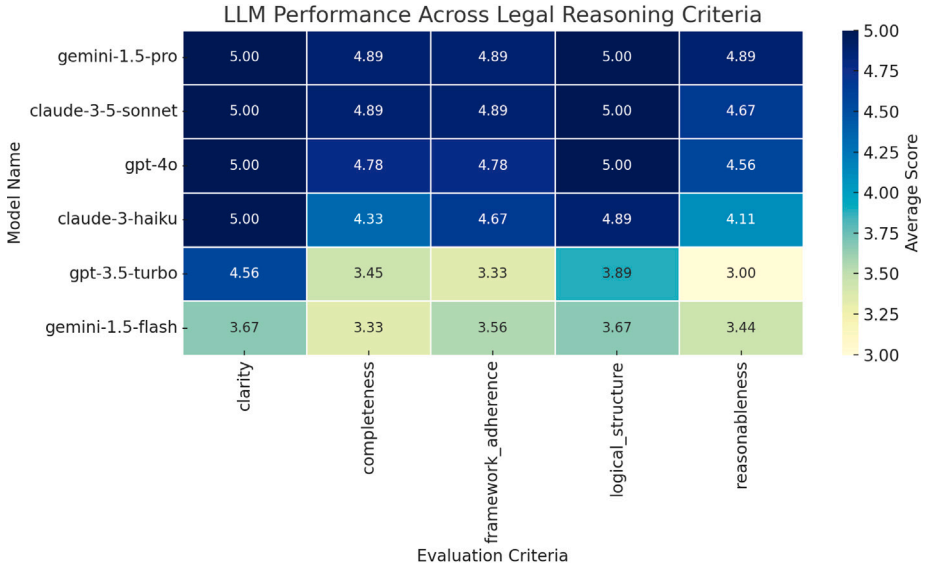


Fig. 5. Average evaluation scores by model across five criteria.

an efficient retrieval system for massive case law data. Moreover, evaluating LLMs is challenging due to the requirement of extensive, intricate training datasets as well as the fundamental tasks involved in curating appropriate those datasets.

6. Open issues and challenges

In practice, legal practitioners may employ various forms of reasoning to make decisions. These reasoning forms can be combinations of rule-based, abductive, and case-based reasoning, and can take broader policy and social factors into account. However, LLMs and other language modeling technologies can only approximate and not fully capture the complexity of real-world legal reasoning.

6.1. Policy-based reasoning

Policy-based reasoning involves analyzing the implications of public policies and policy objectives for interpreting laws and applying them to specific cases [85,86]. This more comprehensive approach might consider the broader societal implications of the legal decision and aim to promote fairness, efficiency, or stability, within current cultural contexts. Lawyers and judges may use policy-based reasoning to determine whether a specific application or interpretation of the law aligns with larger societal goals and to resolve any conflicts between competing legal principles.

6.2. Principled reasoning

While policy-based reasoning focuses on the implications and objectives of public policies in a broad scope, principled reasoning instead relies on underlying normative values, ethical principles, and fundamental justice to inform legal interpretation and decision-making [87]. This approach emphasizes the role of moral and philosophical values in law, acknowledging that legal judgments often rely on subjective beliefs about fairness, equality, and propriety. So both lawyers and judges may invoke principles such as autonomy, dignity, and equity when resolving disputes and interpreting potential vagueness and ambiguities in the law.

6.3. Cultural-based reasoning

In addition to policy aspects of legal decision making identified above, *cultural-based reasoning* takes into account cultural norms, practices, and traditions when interpreting laws and resolving legal disputes [88]. This form of reasoning acknowledges that the law often

reflects and reinforces the current cultural and social beliefs of a specific society at a specific time in history, and then contextualizes the legal decision within this broader framework. Lawyers and judges may therefore turn to prevailing customs and societal expectations as an additional source of legal guidance or validation, particularly when the law is unclear or silent on a particular issue.

6.4. Justice as a source of law and legal reasoning

Justice is a fundamental concept in the realm of law and legal reasoning, representing a core value that underlies the development, interpretation, and application of laws [89]. Justice is often viewed as a shared value among individuals and societies, which is reflected in the legal texts and documents they create. Being able to identify and understand notions of justice in legal discourse is thus essential for legal practitioners when engaging in tasks that require them to interpret laws, solve disputes, and make legal arguments. Having effective tools, including but not restricted to LLMs, to explore these moral and justice-related values can provide invaluable insights for legal reasoning processes [13].

6.5. Exploring ethical values and justice principles

LLMs can be employed to analyze legal texts, detect and explore ethical values [90,91], and identify expressions of justice principles in various legal documents. This may involve training the models to recognize specific keywords, phrases, or patterns that correlate with these values and principles, in order to provide an overview of the various perspectives on justice present within a corpus. Advanced natural language processing techniques, such as sentiment analysis, text classification, and named entity recognition, can be applied to help facilitate this task.

6.6. Risks of hallucination and overreliance in judicial contexts

While LLMs offer promising capabilities for supporting legal reasoning tasks, their integration into judicial or high-stakes decision-making contexts must be approached with caution. A major limitation is the risk of hallucination—where the model generates fluent but factually incorrect or legally invalid outputs. Equally concerning is the tendency for users, especially non-experts, to over-rely on these outputs without sufficient verification or legal grounding. These risks are especially critical given the potential role of LLMs in influencing legal decisions.

Finally, treating LLMs as a unified framework for legal reasoning should not be conflated with assuming they can reliably convert arbitrary inputs into valid outputs across all reasoning types. To responsibly evaluate and improve such systems, it is essential to examine their behavior through the lens of classical reasoning frameworks including rule-based, abductive, case-based, and beyond. This not only requires the discipline to measure their task-specific effectiveness but also helps identify failure modes and systemic risks, thereby guiding future refinement of the overall framework.

7. Conclusion

The integration of LLMs within a unified reasoning framework encompassing rule-based, abductive, and case-based reasoning holds immense promise for advancing legal analysis and decision-making. By combining the strengths of each reasoning approach, this framework enables LLMs to provide support for comprehensive, accurate, and adaptable legal solutions in diverse contexts. However, the realization of this potential requires addressing numerous open issues and challenges, including the exploration of additional forms of legal reasoning, understanding cultural notions of justice, and ensuring the learnability and adaptability of LLMs in legal reasoning tasks. In addition, the evaluation of LLMs poses significant challenges due to the vast and complex datasets involved, as well as the diverse range of basic tasks such as negation, entailment, extraction, retrieval, and ranking, which all require meticulous design. Addressing these challenges will play a critical role in enhancing the power of AI-driven legal analysis tools, empowering legal practitioners with valuable assistance and insights for a wide range of legal situations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Simulating civil law court procedure

To supplement Section 4, this appendix demonstrates prompts and responses when applying example cases and rules in Section 2 to simulate a civil law court procedure, using Gemini-1.5-flash.

A.1. Rule-based reasoning prompt

The following prompt and response demonstrate the application of *Young v. Hitchens* to the theory, following the proof structure outlined in Section 2, within the rule-based reasoning mode of the framework. The response shows that, based on the available facts and legal theory, the defendant is guilty of theft.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is guilty of theft if:

1. the person committed an action to take away the object,
2. the object belonged to another person,
3. the person knew that the object belonged to the other person, and
4. the person had an intention to take away the object

   The rule is not applicable if the purpose of taking away is to prevent dangers.

According to the given rules, please evaluate whether the defendant is guilty of theft.

Response

Let's analyze the situation based on the provided rules:
1. Action to take away the object: The defendant did row his boat into the plaintiff's net, catch the fish, and row out with the fish. This clearly constitutes an action to take away the fish.
2. The object belonged to another person: The fish were caught in the plaintiff's net, indicating they were in the plaintiff's possession and therefore belonged to the plaintiff.
3. Knowledge of ownership: The defendant, by entering the net and taking the fish, must have known that the fish were caught in someone else's net. This implies knowledge of the fish belonging to another person.
4. Intention to take away: The defendant's actions of entering the net, catching the fish, and leaving with them strongly suggest an intention to take away the fish.
5. Prevention of dangers: There is no mention of any danger that the defendant was trying to prevent by taking the fish.
Based on the provided rules, the defendant appears to be guilty of theft. All four conditions for theft are met [...]. There is no evidence of the defendant acting to prevent any danger, which would have been a potential defense.

A.2. Abductive reasoning prompt

The following prompt and response demonstrate the application of *Young v. Hitchens* to the theory within the abductive reasoning mode of the framework. The response shows possible arguments (as hypotheses) that the defendant may not be guilty of theft in this case.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is guilty of theft if:

1. the person committed an action to take away the object,
2. the object belonged to another person,
3. the person knew that the object belonged to the other person, and
4. the person had an intention to take away the object

   The rule is not applicable if the purpose of taking away is to prevent dangers.

Please suggest hypotheses to support why the outcome of the case might be the defendant is not guilty of theft.

Response

Several hypotheses could support a finding of not guilty for the defendant, even if the facts seem to initially suggest theft:
Hypothesis 1: Lack of Knowledge of Ownership (Condition 3)
Argument: The defendant might argue he didn't know the fish belonged to the plaintiff. The open sea is a common fishing ground. The net wasn't clearly marked as belonging to the plaintiff, and the defendant might claim he believed the fish were simply in the water and available for anyone to catch. The act of the net being almost closed might not be sufficient to establish clear ownership in the defendant's eyes, especially in the context of open sea fishing.
Hypothesis 2: Ambiguity of Possession (Condition 2)
Argument: While the plaintiff had a net deployed, the fish weren't definitively "possessed" until they were fully secured within the net. The defendant could argue that the fish were still in the common resource of the open sea until that point, and therefore not yet the property of the plaintiff. The act of the net being almost closed, but not fully closed, creates a grey area regarding possession.
Hypothesis 3: Accidental Taking (Condition 1 & 4)
Argument: The defendant might argue that entering the net and taking the fish was accidental. Perhaps he didn't see the net clearly, or misjudged his boat's position. If the court finds the act of taking the fish was unintentional or accidental, the element of "intention to take away the object" (Condition 4) is not met. This would require evidence that the defendant didn't deliberately target the net or the fish.
Hypothesis 4: Defense of Necessity (Exception to the Rule)
Argument: While unlikely, the defendant might attempt a "prevention of danger" defense (the exception to the rule). This would be a weak argument and require extraordinary circumstances. For example, he might claim that his boat was in imminent danger of collision and entering the net was the only way to avoid a more significant accident. This would need strong supporting evidence.
Hypothesis 5: Insufficient Evidence of Intent (Condition 4)
Argument: Even if the defendant knew the net belonged to someone, the prosecution might not be able to prove beyond a reasonable doubt that he *intended* to steal the fish. revised rule thought

A.3. Rule-based reasoning prompt (with new hypothesis)

The following prompt and response demonstrate the application of *Young v. Hitchens* to the theory within the rule-based reasoning mode. This application is now with a new hypothesis generated from the previous prompt and accepted by the judge. The hypothesis states that "the fish were still in the common resource of the open sea until that point, and therefore not yet the property of the plaintiff". The response shows that, with the new hypothesis, the defendant is now not guilty of theft.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is guilty of theft if:

1. the person committed an action to take away the object,
2. the object belonged to another person,
3. the person knew that the object belonged to the other person, and
4. the person had an intention to take away the object

   The rule is not applicable if the purpose of taking away is to prevent dangers.

Assuming these hypotheses are true in this case:

The fish were still in the common resource of the open sea until that point, and therefore not yet the property of the plaintiff.

According to the given rules and hypotheses, please evaluate whether the defendant is guilty of theft.

Response

Based on the provided rules and hypotheses, the defendant is *not guilty of theft*. Here's why:
While the defendant undeniably took the fish (condition 1), and he did so with the intention of keeping them (condition 4), the crucial element missing is condition 2: the fish did not yet belong to the plaintiff. The hypothesis explicitly states the fish were still in the common resource of the open sea and therefore not yet the property of the plaintiff. Until the plaintiff's net was fully closed and the fish were effectively secured, they remained unowned. The defendant's act, therefore, did not involve taking something that belonged to another. The act of taking the fish from the net before it was fully closed does not automatically transfer ownership.

Appendix B. Simulating common law court procedure

To supplement Section 4, this appendix demonstrates prompts and responses when applying example cases in Section 2 to simulate a common law court procedure, using Gemini-1.5-flash.

B.1. Case-based reasoning prompt

The following prompts and responses demonstrate the arguments of *Young v. Hitchens* from each side using the case-based reasoning mode of the framework. The plaintiff uses *Keeble v. Hickeringill* and the defendant uses *Pierson v. Post* to support their arguments.

Plaintiff side

Prompt

Considering the current case:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the precedent case:

1. The plaintiff owned land containing a pond, where the plaintiff always takes ducks for a profit.
2. The defendant, knowing about the decoy pond, fired the guns to scare away the ducks in the pond.

The precedent case has the following outcome: the defendant is guilty of theft

Please answer why the current case might have the same outcome in the following form and make a rule-based explanation between the current case and the rule of that form:

A person is typically guilty of theft if the following conditions hold:

1. ...
2. ...

Response

A person is typically guilty of theft if the following conditions hold:

1. They intentionally and without consent take possession of another person's property.
2. That property has economic value to the owner and is considered personal property.

Rule-Based Explanation:
Both cases involve the intentional deprivation of another's property for economic gain. The precedent case establishes that interfering with the process of acquiring property (the ducks) that is already under the plaintiff's control and intended for profit constitutes theft. The current case mirrors this: the plaintiff had effectively possessed the fish (the net was almost closed, signifying imminent capture and control), and the defendant's actions directly interfered with this possession, depriving the plaintiff of the fish and their economic value. The fish, like the ducks, were in the process of being lawfully acquired and represented a form of personal property with economic value. The defendant's intentional act of entering the net and taking the fish, without the plaintiff's consent, satisfies the conditions for theft. The location (open sea vs. pond) is irrelevant to the core issue of intentional deprivation of property under the plaintiff's control.

Defendant side

Prompt

Considering the current case:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the precedent case:

1. The plaintiff chased a fox in an open land.
2. The defendant killed it and carried it away.

The precedent case has the following outcome: the defendant is not guilty of theft

Please answer why the current case might have the same outcome in the following form and make a rule-based explanation between the current case and the rule of that form:

A person is typically guilty of theft if the following conditions hold:

1. ...
2. ...

Response

A person is typically guilty of theft if the following conditions hold:

1. They unlawfully appropriate property belonging to another.
2. That property is considered tangible and has been reduced to possession by the owner.

Rule-Based Explanation:
The precedent case (fox hunting) and the current case (fishing) share a key similarity: *lack of clear, established possession*. In the fox hunting case, the fox, being a wild animal, was not considered the property of the pursuer until it was captured. Similarly, in the fishing case, while the plaintiff had a net around the fish, the fish were not yet securely in his possession. The net was *almost* closed, implying the fish were still in the open sea and not definitively under the plaintiff's control. The defendant's act, while arguably opportunistic and unfair, didn't involve taking something already clearly owned by the plaintiff. Both cases involve the taking of something not yet fully possessed by the claimant. Therefore, the argument could be made that neither the fox nor the fish were "property belonging to another" in the legal sense required for theft at the moment of appropriation. The rule above highlights this crucial element of possession as a prerequisite for theft.

B.2. Abductive reasoning prompt

The following prompts and responses demonstrate the challenges in the case of *Young v. Hitchens* from the opposing side using the abductive reasoning mode of the framework.

B.2.1. Defendant challenges plaintiff's theory

The response shows that "lack of possession" or "lack of intention" can be counterarguments that the defendant can use with the plaintiff.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is typically guilty of theft if the following conditions hold:

1. They intentionally and without consent take possession of another person's property.
2. That property has economic value to the owner and is considered personal property.

Please suggest hypotheses to support why the outcome of the case might be the defendant is not guilty of theft.

Response

Several hypotheses could support a finding that the defendant is not guilty of theft, based on the provided rule:

Hypotheses focusing on the lack of "possession" by the plaintiff:

1. *The fish were not yet in the plaintiff's possession:* The hypothesis argues that merely having a net around fish in the open sea doesn't constitute possession. The fish were still in the open sea, and the plaintiff hadn't yet physically secured them. The act of closing the net was incomplete, leaving the fish arguably still unowned. The defendant merely took advantage of an opportunity before the plaintiff established clear possession.
2. *The plaintiff lacked sufficient control over the fish:* This hypothesis builds on the first. Even with the net partially closed, the plaintiff lacked complete control over the fish. The open-sea environment allowed for interference, and the defendant's actions, while arguably disruptive, didn't necessarily violate the plaintiff's existing, incomplete control.

Hypotheses focusing on the lack of "intentional taking of another's property":

3. *The defendant lacked knowledge of the plaintiff's claim of possession:* The defendant might argue he didn't know the net belonged to the plaintiff or that the plaintiff was actively trying to catch the fish. If the defendant believed the fish were unowned, the element of intentionally taking *another's* property is missing. This would require evidence about the visibility of the net and the plaintiff's actions.
4. *The defendant's actions were unintentional or accidental:* The defendant might argue his entry into the net was accidental, perhaps due to poor visibility or unforeseen circumstances. If the taking of the fish was unintentional, the element of intentional taking is not met.

[...]

B.2.2. Plaintiff challenges defendant's theory

The response shows that "direct intention" or "economic value of the fish" can be counterarguments that the plaintiff can use with the defendant.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is typically guilty of theft if the following conditions hold:

1. They unlawfully appropriate property belonging to another.
2. That property is considered tangible and has been reduced to possession by the owner.

Please suggest hypotheses to support why the outcome of the case might be the defendant is guilty of theft.

Response

To argue that the defendant is guilty of theft, we need to show that both conditions of the rule are met. Here are some hypotheses supporting guilt, focusing on how each condition might be satisfied:

Hypotheses supporting Condition 1: Intentional and without consent taking of possession:

Hypothesis 1a (Direct Intention): The defendant intentionally entered the plaintiff's net, knowing it contained fish belonging to the plaintiff, and intentionally took those fish. His actions demonstrate a clear intent to deprive the plaintiff of his property. The act of rowing into the net and removing the fish wasn't accidental; it was a deliberate act of appropriation.
Hypothesis 1b (Reckless Disregard): The defendant acted with reckless disregard for the plaintiff's property rights. Even if he didn't explicitly intend to steal, his actions of rowing into a clearly visible and actively used fishing net, knowing it likely contained fish, demonstrate a reckless disregard for the possibility of harming the plaintiff's possession. This reckless disregard could be considered equivalent to intentional action in a legal context.

[...]

Hypotheses supporting Condition 2: Property has economic value and is considered personal property:

Hypothesis 2a (Economic Value): The fish caught in the net had economic value to the plaintiff. This is self-evident; fish are commonly bought and sold, and the plaintiff was actively engaged in fishing, implying an intention to sell or consume the fish for profit or sustenance.
Hypothesis 2b (Personal Property): The fish, once caught in the net, became the personal property of the plaintiff. The act of fishing, using a net to capture the fish, establishes a claim of ownership. The fish were no longer wild and freely available; they were under the plaintiff's control and in the process of being secured.

[...]

B.3. Rule-based reasoning prompt (with new hypothesis)

The following prompt and response demonstrate the application of *Young v. Hitchens* to the plaintiff's theory within the rule-based reasoning mode. The application is with a new hypothesis generated from the defendant's challenge and accepted by the judge. Since the plaintiff's challenge and the defendant's challenge are opposite to each other, it is under the discretion of the judge to choose to accept which challenge as the hypothesis. In this example, we follow the real case in which the judge accepts the defendant's hypothesis that "the fish were not yet in the plaintiff's possession as the fish were still in the open sea and the plaintiff hadn't yet physically secured them". The response shows that the defendant successfully challenges the plaintiff's theory with the new hypothesis.

Prompt

Considering the following facts:

1. The plaintiff was fishing with a net in an open sea.
2. While the net was almost closed, the defendant rowed his boat, entered the net, caught the fish, and rowed his boat out of the net with the fish.

Considering the following rules: A person is typically guilty of theft if the following conditions hold:

1. They intentionally and without consent take possession of another person's property.
2. That property has economic value to the owner and is considered personal property.

Assuming these hypotheses are true in this case:

The fish were not yet in the plaintiff 's possession as the fish were still in the open sea and the plaintiff hadn't yet physically secured them.

According to the given rules and hypotheses, please evaluate whether the defendant is guilty of theft.

Response

Based on the provided rules and hypotheses, the defendant is likely *not* guilty of theft.
The crucial point is the hypothesis: "The fish were not yet in the plaintiff's possession as the fish were still in the open sea and the plaintiff hadn't yet physically secured them."

For theft to occur, the defendant must have taken possession of *another person's property*. While the plaintiff was *attempting* to catch the fish, they hadn't yet established possession. The fish were still in the open sea, a common resource. The plaintiff's net, while deployed, hadn't yet effectively enclosed and secured the fish. Therefore, the fish weren't legally considered the plaintiff's property at the moment the defendant intervened.

The defendant's actions might be considered unethical or even a civil trespass, but they don't meet the legal definition of theft as presented. The defendant didn't take possession of something already belonging to the plaintiff.

Data availability

Data will be made available on request.

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